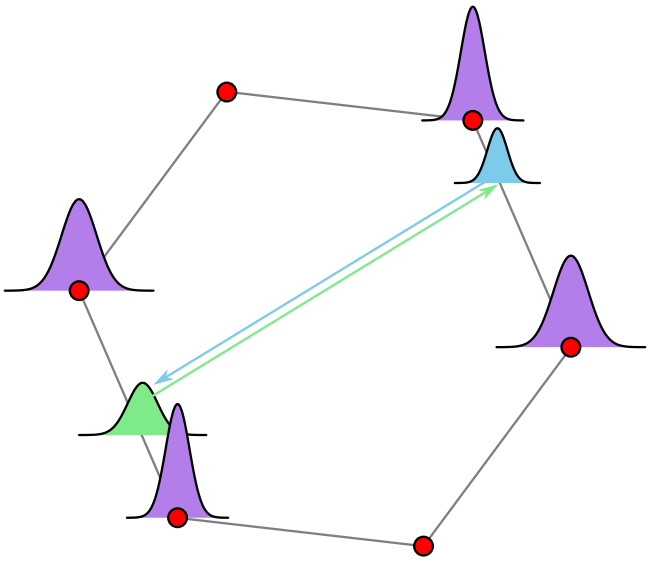


Final Report

The Art of Scientific Computing: Project Title: A selection of granary queueing models

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Chapter 1

Introduction

In which we talk about granaries and justify the importance of the question

Granaries are storage structure for grain, used to protect harvested grain from poor weather and pests. To prevent spoilage grain needs to kept at a low moisture content; this state may initially be achieved by harvesting during dry conditions, or by drying harvested grain. Drying sometimes happens at the granary.

It is typical for a granary to service all of the farms in an area, often operating as either a collective or through commercial agreements.

Grain must be harvested at the correct time of year, and the need for dry weather leads to busy peak periods for granaries, corresponding with appropriate weather. Deliveries can cluster, as farmers restart harvesting operations after inclement weather or race to harvest as much as possible before rain.

These high demand periods can lead to significant queues, incurring costs of time and money. Understanding the likely demands placed on a granary is therefore important to decisions regarding its design and operation. Planners may wish to invest in equipment to reduce unloading time or add additional service bays; managers may wish to alter policy regarding opening hours or the order in which trucks are served.

Mathematical models are a useful way of investigating these demands and evaluating different investments and policies. In this investigation we implement three different models of a granary, and investigate where the models agree and disagree. We compare to closed form results for queueing systems, and consider where the models likely fall short of reality.

Chapter 2

Background

In which we cover basic queueing theory, classify our problem and mention the closed form results that are going to be important for verification.

Chapter 3

Models

In which we discuss the three models of a granary we're going to be using

First the shared infrastructure of a simple queue and a single deterministic server

- An open loop with Markov modulated Poisson arrival
- A closed loop with an abstracted truck fleet and an abstracted grain supply
- A simulation of all the farms in the catchment each with their own trucks

Chapter 4

Implementation

In which we discuss the implementation in python. Discuss the initial simpy profiling, and the subsequent handmade simulation with a queue using a deque and a priority queue of upcoming events using a heapq. Discuss parallelisation with joblib - embarassingly parallel

Chapter 5

Validation

Out come the closed form results, most useful for model 1. In correct regimes model 2 and 3 should converge to something like those results as well

Chapter 6

Experiments

How do mean and tail waiting times change as we increase weather variance?
- How does overtime increase as we increase weather variance?
- How do the models compare to one another?
- Report on parallelisation efficiency

Chapter 7

Results and Discussion

When models are tuned to be similar, do the tunings make sense analytically?
Where do models disagree when tuned to be similar?

Chapter 8

Conclusion